L-F-00320D (GSA-FSS)
August 21, 1975
SUPERSEDING
Int. Fed. Spec. L-F-00320C (GSA-FSS)
November 17, 1972, and
INTERIM REVISION OF
Fed. Spec. L-F-320B
May 4, 1965

INTERIM FEDERAL SPECIFICATION

# FILM, THERMAL DEVELOPING, VESICULAR

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification L-F-320B dated May 4, 1965.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for thermal developing vesicular negative-working duplicating film supplied in rolls and in sheets in sizes normally used in microreproduction systems. The duplicating film specified herein is of a type in which the image is composed of light-scattering vesicles in a plastic vehicle on a flexible transparent base.

1.2 Classification.

1.2.1 Types, subtypes, classes and kinds. The thermal developing film covered by this specification shall be of the following type, subtypes, classes and kinds.

Type:

I - Duplicating

Subtypes:

A - Sign-reversing

Classes:

1 - High Speed

Kinds:

1A - 2.5 mil
1B - 3 mil
1C - 4 mil
1D - 5 mil
1E - 7 mil

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-T-60 - Tape, Packaging, Waterproof.

FSC 6750

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Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MS-26565 - Spool - Photographic Film, Aerial.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute, Inc. (ANSI) Publications:

PH1.13 - Cores for Photographic Film Rolls (Plastic, Wood, or Metal), Dimensions for.
PH1.20 - 70 mm Unperforated and Perforated Film for Camera other than Motion-Picture Cameras, Dimensions for. (ISO R193)
PH1.25 - Safety Photographic Film, Specifications for. (ISO R543)
PH1.33 - 16 mm 100-Foot, 16 mm 200-Foot, 35 mm 100-Foot, and 70 mm 100-Foot Spools for Recording Instruments and for Microfilm and Still-Picture Cameras, Dimensions for.
PH2.2 - Sensitometry of Photographic Papers.
PH5.3 - 16 mm and 35 mm Silver Gelatin Microfilms for Reel Applications, Specifications for. (ISO R193 and R260)

(Copies may be obtained from the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

American Society for Testing and Materials (ASTM) Standards:

D-882-56T - Methods of Tests for Properties of Thin Plastic Sheets and Films.
D-1004-49T - Methods of Test for Initial Tear Strength.

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)



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Technical Association of Pulp and Paper Industry (TAPPI) Standards:

T411os-68 - Thickness (Caliper) of Paper and Paperboard.

(Copies may be obtained from the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30341.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials. Film procured in accordance with this specification shall meet all the requirements herein specified. Any film which exhibits coating or physical defects not meeting the specification shall be replaced by the manufacturer at no additional cost to the government.

3.1.1 Film base. The film base shall be a polyester film which is an optically homogenous, clear, transparent medium and shall be noncolor selective and supplied perforated or nonperforated as specified.

3.1.1.1 Perforations. Unless otherwise specified, perforated film shall have type II perforations in accordance with ANSI PH1.20.

3.1.1.2 Safety base. The base material shall meet the safety limits as defined in ANSI PH1.25, "Safety Photographic Film."

3.1.2 Spools and cores. Unless otherwise specified all roll film specified herein shall be supplied on standard spools or cores. For 16 mm and 35 mm film widths, the spools shall conform to ANSI PH1.33. For 70 mm and larger film widths the spools shall conform to MS-26565. Cores, when specified, shall conform to ANSI PH1.13.

3.1.3 Tensile strength. The tensile strength of the base film shall not be less than 10,000 p.s.i. when tested as specified in 4.3.2.

3.1.3.1 Initial tear strength. Initial tear strength of the base film shall not be less than 1,000 pounds per inch of thickness when tested as specified in 4.3.3.

3.1.4 Roll film.

3.1.4.1 Widths. Roll film shall be slit in accordance with table I. Dimensions shown apply to the material immediately after slitting.

TABLE I - Slitting widths and tolerances

| Nominal width | Film width and tolerances | |
|---|---|---|
| | Metric (millimeters) | Inches |
| 16mm | 15.95 + 0.025/-0.050 | 0.628 + 0.001/-0.002 |
| 35mm | 34.975 + 0.025/-0.050 | 1.377 + 0.001/-0.002 |
| 70mm | 69.95 ± 0.05 | 2.754 ± 0.002 |
| 105 mm | 104.87 ± 0.13 | 4.129 ± 0.005 |
| 3-1/4 inch | 82.42 ± 0.13 | 3.245 ± 0.005 |
| 5 inch | 125.98 ± 0.25 | 4.960 ± 0.010 |
| 9-1/2 inch | 240.28 ± 0.25 | 9.460 ± 0.010 |



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3.1.4.2 Lengths. The film lengths shall be as follows: 400, 475, 500, 1000, and 2000 feet. The total length furnished shall include both the film length and the combined integral film leader and trailer not less than 15 feet in length for all combinations of length and width.

3.1.5 Sheet film. Sheet film dimensions shall be in accordance with table II.

TABLE II - Sheet film dimensions

| Nominal dimensions | Metric (millimeters) | Inches |
|---|---|---|
| 105 mm x 148 mm | 105.00 + 0.00/-0.75<br>x 148.00 + 0.00/-1.00 | 4.134 + 0.000/-0.030<br>x 5.827 + 0.000/-0.039 |
| 3-1/4 in. x 7-3/8 in. | 82.55 + 0.18/-0.00<br>x 187.33 ± 0.127 | 3.250 + 0.007/-0.000<br>x 7.375 ± 0.005 |

3.1.6 Thickness. The thickness of the film before processing with coating included shall be as given in table III when measured using the method given in 4.3.4.

TABLE III - Film thickness

| Kind | Base thickness plus coating (inches) |
|---|---|
| 1A | 0.0024 ± 0.0005 |
| 1B | 0.0034 ± 0.0007 |
| 1C | 0.0044 ± 0.0008 |
| 1D | 0.0054 ± 0.0010 |
| 1E | 0.0074 ± 0.0010 |

3.1.7 Coating defects. The film shall be free from scratches, pinholes, blisters, tears, fingerprints, or other physical defects; and from streaks, spots, or areas of marked higher or lower sensitivity than surrounding portions when tested as specified in 4.3.5.

3.1.8 Coating adhesion. The coating adhesion of the film specified herein shall comply with paragraph 4.3.6.

3.2 Sensitometric characteristics. All films shall have sensitometric characteristics in conformance with table IV.

3.2.1 Density. All densities specified herein shall be projection densities as defined herein. The maximum density shall be not less than that shown in table IV when tested as specified in 4.3.10.1. The minimum density shall not exceed that shown in table IV when film is exposed and processed in accordance with 4.3.10.2.

3.2.2 Bar gamma. The bar gamma shall be as specified in table IV, and shall be measured by the method defined in ANSI PH2.2 as specified in 4.3.10.4.

TABLE IV - Sensitometric characteristics

| Type | Subtype | Class | D Max.* (min) | D Min. (max) | Bar Gamma (min) | Speed-rating at** net projection density of 1.8 | Resolution lines/mm (minimum) |
|---|---|---|---|---|---|---|---|
| 1 | A | 1 | 2.20 | 0.35 | 2.0 | 0.65 to 0.95 | 161 |

* Density scale shall not be less than 1.90.
** Speed variations among lots of the same brand and product designation (having different emulsion numbers) shall not exceed ± 0.10, and in no instance shall the speed-rating of the lot average fall outside the class range. Within a given lot (having a specified emulsion number), speed variations shall not exceed ± 0.075, and in no instance shall the speed-rating fall outside the class range.

3.2.3 Resolution. Resolution of all film specified herein shall be not less than the lines per millimeter shown in table IV, when tested as specified in 4.3.11.



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3.2.4 Speed rating. Speed rating shall meet the values shown in table IV when tested as specified in 4.3.8 through 4.3.10.3.

3.2.5 Thermal resistance characteristics.

3.2.5.1 Speed change - unprocessed film. When tested as specified in 4.3.12.1, the loss in speed rating shall not exceed 0.15.

3.2.5.2 Density change - processed film. When tested as specified in 4.3.12.2, the loss in density of the step nearest the speed point, i.e., nearest 1.80 plus minimum density, shall not exceed 25%.

3.3 Expiration date. Expiration date shall be not less than two years from date of shipment when the film is stored in the original unopened packages in a room maintained at a temperature not higher than 73° F and 50% relative humidity. The expiration date shall be shown on the package.

3.4 Uniformity of characteristics. Uniformity of physical and sensitometric characteristics among specimens of the same manufacture and emulsion number, and among different manufacturing lots of a given type, class, and speed shall be maintained within closest practical limits.

3.5 Workmanship. Workmanship shall be in accordance with high grade commercial practice covering the manufacture of vesicular film by coating polyester base with light sensitive elements in a plastic vehicle. The film shall be completely suitable for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter. For purposes of sampling, an inspection lot for examination and tests shall consist of all materials of the same type and class submitted for delivery at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.2.2 Inspection of the end item.

4.2.2.1 Examination of the end item. Examination of the end item shall be made in accordance with the classification of defects, set forth below and inspection levels and acceptable quality levels (AQLs) specified in 4.2.2.1.5. Not more than 3 rolls of roll film shall be drawn from any one shipping container for each lot of film offered for acceptance for visual and dimensional characteristics. The lot size for purposes of determining sample size in accordance with MIL-STD-105 shall be expressed in units of rolls for examination under 4.2.2.1.1, 4.2.2.1.2, and 4.2.2.1.3 and in units of shipping containers for examination in 4.2.2.1.4. Examinations included in 4.2.2.1.1, 4.2.2.1.2, and 4.2.2.1.3 may be performed at any point immediately before, during or after the spooling or packaging operation.

4.2.2.1.1 Examination of the end item for defects in appearance and workmanship. The sample unit for this examination shall be one linear foot of sensitized film as applicable. Not more than three sample units shall be drawn from any one roll.



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| Examine | Defect |
|---|---|
| Appearance and workmanship | Film not completely or evenly coated. |
| | Emulsion surface shows scuffs, scratches, foreign matter, or abrasions. |
| | Holes, pinholes, tears, cuts, cracks, fingermarks, streaks, spots, or sharp creases. |

4.2.2.1.2 Examination of the end item for dimensional defects. The sample unit for this examination shall be one roll of film or one linear foot of film as applicable.

| Examine | Defect |
|---|---|
| Width | Not within tolerances. |
| Thickness | Not within tolerances. |

4.2.2.1.3 Examination of the end item for defects in construction and length and blocking. The sample unit for this examination shall be one roll of roll film.

| Examine | Defect |
|---|---|
| Roll | Spool (when required) not specified type, damaged. Film unevenly or improperly wound; less than specified number of feet. Roll not continuous. Roll contains splice. |

4.2.2.1.4 Examination of preparation for delivery. An examination will be made to determine that marking, materials, workmanship, contents and weights comply with the requirement of section 5 of this specification. Defects will be scored in accordance with the classification below. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

| Examine | Defect |
|---|---|
| Packaging | Not as specified, not in accordance with contract requirements. Rolls not packaged and sealed as specified. Packages not uniform. |
| Packing | Not packed as specified. Packing material not as specified; closures not accomplished by specified or required methods or materials. |
| | Number of rolls per box less than indicated or specified. |
| Markings (interior and exterior) | Marking not in accordance with contract specifications. Labels when used not properly attached. Markings illegible, incorrect, incomplete, omitted, or not in accordance with contract requirements. |
| Weight (interior and exterior) | Gross weight exceeds contract requirements. |



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4.2.2.1.5 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels, for determining the sample size, and the acceptable quality levels (AQLs), expressed in defects per 100 units, shall be as follows:

| Examination | Inspection Level | AQL |
|---|---|---|
| 4.2.2.1.1 | I | 1.0 |
| 4.2.2.1.2 | S-3 | 2.5 |
| 4.2.2.1.3 | S-2 | 2.5 |
| 4.2.2.1.4 | S-2 | 4.0 |

4.2.3 Testing of the end item. Testing of the end item shall be conducted in accordance with table V for the characteristics as indicated therein on each lot submitted for acceptance. The sample unit, as referenced herein is defined as the amount of material required to perform all tests one time. The sample unit for all testing shall consist of one roll of film. Three sample units, each drawn from a different shipping container, shall be tested with no evidence of failure to meet sample unit or average requirements, as applicable. As an alternate, samples for testing for base density may be drawn at any point after the base casting or extrusion operation. Samples for testing for the sensitometric characteristics, may be drawn at any point after the emulsion coating operation.

TABLE V - Instructions for testing (sample unit)

| Characteristic | Specification reference | | Rqmts. appl to indiv. unit | Number determinations per sample unit | Results reported as- | |
|---|---|---|---|---|---|---|
| | Requirement | Test method | | | Pass or fail | Numerically to nearest |
| Resolution | 3.2.3 | 4.3.11 | X | Avg. 5 | X | - |
| Sensitometric properties | 3.2 (Table IV) | 4.3.10 | X | Avg. 3 | - | 0.01 |
| Thermal resistance | 3.2.5.1<br>3.2.5.2 | 4.3.12.1<br>4.3.12.2 | X<br>X | Avg. 3<br>Avg. 2 | -<br>- | 0.01<br>1 percent |

4.3 Test methods.

4.3.1 Physical tests The following tests shall be made in accordance with the applicable test methods specified in the American Society for Testing and Materials, and the Technical Association of the Pulp and Paper Industry.

| Test | Method No. |
|---|---|
| Tensile strength | D-882-56T |
| Initial tear strength | D-1004-49T |
| Thickness | TAPPI-T411os-68 |

4.3.2 Tensile strength. Tensile strength shall be tested in accordance with ASTM method D-882-56T, or the latest revision, when the material is at 75°F.

4.3.3 Initial tear strength. Initial tear strength shall be tested in accordance with ASTM method D-1004-49T or latest revision.



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4.3.4 Thickness. Thickness of the material shall be measured using the methods specified in TAPPI-T411os-68.

4.3.5 Coating defects. Two samples of each kind of film to be tested, not less than 12 inches in length, shall be selected at random. For each film one sample shall be selected at random. For each film one sample shall be exposed to a density $\geq$ 1.80 net and processed on applicable equipment. Then the sample shall be held not more than six feet away from an unshielded 60 watt incandescent lamp and visually examined under the light for compliance with the requirements specified in 3.1.7. The other sample shall be examined directly in visible light to determine conformance with the requirements specified in 3.1.7.

4.3.6 Coating adhesion. Not less than five samples of the sensitized material under test shall be fully processed in accordance with the manufacturer's recommendation. The coatings of the processed samples shall not be removed when a one-half inch wide strip of transparent pressure sensitive tape, conforming with PPP-T-60, is placed firmly on the surface avoiding the edge of the film and pulled away with a sharp tug in a direction 90° from the flat plane of the sample under test.

4.3.7 Apparatus for determining sensitometric (D log E) curve. The following apparatus or other equivalent equipment shall be used in determining the sensitometric curve of the film under test. The apparatus is obtainable at the corresponding listed sources:

1. Kalvar Sensitometer Model 1001 B
   SDL Division
   Kalvar Corporation
   909 South Broad Street
   New Orleans, Louisiana 70125

2. International Light Research Photometer
   Model 600

3. Photosensor Model PT200C

4. Filters ND3 and NB 405
   Calibrated with Xenon Lamp and Double Monochromator
   International Light, Inc.
   Dexter Industrial Green
   Newburyport, Massachusetts 01950

5. Eastman Carbon Step Wedge, No. 5,
   Calibrated
   Eastman Kodak
   Rochester, New York

6. Kalvar Developer, Model No. 7
   Kalvar Corporation
   909 South Broad Street
   New Orleans, Louisiana 70125

7. Kalvar-Baus Primary f/4.5
   Projection Densitometer
   Kalvar Corporation
   909 South Broad Street
   New Orleans, Louisiana 70125

8. Secondary f/4.5 Projection Densitometer
   Kollmorgen Corporation
   MacBeth Division
   P.O. Box 950
   Newburgh, N.Y. 12550

   Sargent-Welch Scientific Company
   7300 North Linder Avenue
   Skokie, Illinois 60076



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9. Temperature Tape
Paper Thermometer Company
10 Stagg Drive
Natick, Massachusetts

10. U.S.A.F. Resolution Target
High Contrast 228 L/mm
Itek Business Products
1001 Jefferson Road
Rochester, New York 14603

11. Kalvar 400 watt Sensitometer Lamp Model #1
SDL Division
Kalvar Corporation
909 South Broad Street
New Orleans, Louisiana 70125

12. Primary Calibrated Wedges of
Commercial Vesicular Films
Kalvar Corporation
909 South Broad Street
New Orleans, Louisiana 70125

Association of Reproduction
Materials Manufacturers (ARMM)
901 North Washington Street
Alexandria, Virginia 22214

4.3.8 Sensitometer calibration. The sensitometer shall be calibrated by measuring the actinic energy density at the sensitometer film plane. Using the IL 600 Photometer, Photosensor PT200C with filters QND3 and NB405, center the Photosensor over the shutter of the sensitometer with the lamp on and shutter open, record the reading on the IL 600 Photometer. Compare the reading to the calibration value supplied by the manufacturer. If the reading is different, vary the voltage to the sensitometer lamp until the exposing energy is equal to 0.0084 watts/$cm^2$. (If Dmax is not achieved in this exposure, then expose a separate film sample to an energy of 0.07 watt-sec/$cm^2$ or more to establish Dmax.) Wait five minutes and repeat until there is no change in the Photometer reading on repeated measurement. Record the reading on the sensitometer. Each time the sensitometer is turned on, wait ten minutes, open shutter and adjust the lamp voltage to give the correct meter reading as required.

4.3.9 Secondary f/4.5 projection densitometer calibration. Secondary f/4.5 projection densitometers shall be prepared by calibration using a vesicular step tablet calibrated by the Kalvar-Baus or equivalent absolute densitometer. The secondary projection densitometer shall be equipped with a Wratten 106 filter and and shall be considered as calibrated only for that type and brand of film used for its calibration. For each other type and brand of vesicular film, a primary calibrated wedge of this type and brand may be obtained from Kalvar Corporation to confirm that the secondary projection densitometer readings for this type and brand agree with the calibration values. Calibrated wedges are also available from ARMM. If the readings are not in accord with the primary calibration values, a calibration curve shall be prepared relating readings on the secondary projection densitometer to calibration projection density values for the film of other types and brands.

4.3.10 Preparation of D log E curve. Turn on the sensitometer, Kalvar Developer, and the f/4.5 projection densitometer. Wait fifteen minutes. Check the open shutter reading on the sensitometer ammeter and adjust to an exposure energy equal to 0.0084 watts/$cm^2$ with powerstat. Close the shutter. Adjust temperature of developer to within + 5° F. of value recommended by the manufacturer. Tape the Kodak density strip on the sensitometer, emulsion side up. Place the test film strip, emulsion



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down, in contact with the carbon density strip and fasten down the contact belt. Open the shutter. Expose for 4.0 seconds. The film temperature shall not exceed 100° F during the exposure as indicated by a suitable temperature sensor. When the shutter closes, remove the test film strip immediately and develop within five seconds, in the Kalvar Developer. Clear the exposed and developed test film strip according to manufacturers' recommendations. Measure the densities in the center of each of the various steps in the resulting step tablet using the Kalvar-Baus primary projection densitometer or a suitably calibrated secondary projection densitometer. Plot a sensitometric curve (D log E) from the measured densities on the vertical axis against the calibrated densities of the Kodak density strip on the horizontal axis. Connect the plotted points with a smooth line thus producing the sensitometric curve. All tests shall be performed under gold fluorescent safelights or equivalent safelights.

4.3.10.1 Maximum projection density. The maximum projection density shall be measured on a sample piece of film which is first exposed on a calibrated Kalvar Standard Sensitometer, Model B or a sensitometer of equal or better accuracy. The sample shall be exposed to 0.0089 watts/$cm^2$ or greater of actinic energy for a time not to exceed 10 seconds and then processed within 5 seconds in accordance with the manufacturer's instructions. The film temperature shall not exceed 115° F during exposure as indicated by using temperature sensing paper or any other suitable means.

4.3.10.2 Minimum projection density. The lowest density value measured on the step tablet generated in 4.3.10 is the minimum projection density (D-min).

4.3.10.3 Speed rating On the vertical axis of the graph generated in 4.3.10, find the value equivalent to the sum of 1.80 plus the minimum projection density. Extend a horizontal line from this value to the D log E curve. From this point drop a vertical line intercepting the horizontal axis. This log E value is the Speed Rating of the sample of film. Repeat this procedure with two more samples. Average the results and record this average as the Speed Rating of the film.

4.3.10.3.1 Speed Rating; alternate method. The following alternate method is provided for the user as a means of obtaining an approximate speed rating for a given film and is not intended for use in qualifying a given film product since the errors involved can be as great as ± 0.10. This alternate method of determining the speed rating of a vesicular film consists of comparing its speed to that of a known standard vesicular film material. Such a known standard is the ARMM (Association of Reproduction Materials Manufacturers) Vesicular Microfilm Speed Standard and is available from ARMM, 901 North Washington Street, Alexandria, Virginia 22214. Under this alternate method expose both the ARMM Vesicular Microfilm Speed Standard and the film whose speed is to be determined simultaneously by equal sharing of an Eastman carbon stepwedge #5, to the ultraviolet lamp in a commercial ultraviolet microfilm printer. The actinic light intensity must be sufficiently great that an f/4.5 projection density of at least 1.80 plus Dmin is obtained on both films in an exposure time not to exceed 10 seconds. Develop the two films within 5 seconds in a vesicular developer according to the manufacturer's recommendations. Measure the densities of the various steps in the resulting step tablets using the Kalvar-Baus Primary Projection Densitometer or a suitably calibrated secondary projection densitometer. Plot a sensitometric curve (D log E) for each film on the same sheet of graph paper using the method described at the end of paragraph 4.3.10. For each D log E curve find the point on the curve equivalent to a density of 1.80 + Dmin. From each of these points drop a vertical line intercepting the horizontal axis. If the curve for the test film sample is to the left of the speed standard then the test film is faster than the standard film by an amount equal to the difference between the intercepts on the horizontal axis. If the curve for the test film is to the right of the curve for the standard film then the test film is slower than the standard film by an amount equal to the difference between the intercepts on the horizontal axis. Repeat this procedure with two more samples of both the standard film and the test film. Average the results. Add or subtract the average to the speed rating of the standard film to obtain the approximate speed rating of the test film.



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4.3.10.4 Bar gamma. The bar gamma shall be determined from the sensitometric curve generated in 4.3.10 by the method defined in ANSI PH2.2.

4.3.11 Resolution. Air Force 1951 High Contrast Positive or Negative Silver Resolution Target or equivalent shall be contact printed onto the film under test using a vacuum printing frame or an equivalent spring-loaded printing frame. Exposure shall be made using any highly collimated ultraviolet light source such as an Illumination Industries Incorporated 100 watt, high-pressure, Mercury Vapor lamp (Model III-110) contained in an Illumination Industries lamp house (Model LH-350) with suitable power supply. The film may conveniently be 6 to 12 inches from the lamp house and an exposure sufficient to give a sample density greater than 1.50 shall be used with an exposure time not to exceed 10 seconds. The temperature of the film during exposure should not exceed 115° F. The film being tested should be examined for resolution using an 80 - 100 power microscope, and the last group resolved shall be that group in which it is possible to count the individual lines in either direction.

4.3.12 Thermal resistance characteristics.

4.3.12.1 Speed change - unprocessed film. A sealed production package of roll film or sheet film shall be placed in a 120°F oven for a period of 15 hours. The package shall then be removed from the oven and allowed to come to room temperature. Measure the speed of the film using the procedure in paragraphs 4.3.8 through 4.3.10.3. The difference between the speed rating obtained on the heat-treated film and the speed rating obtained on the film before heat-treating shall be recorded as the change of speed of unprocessed film.

4.3.12.2 Density change - processed film. Prepare two step tablets from the film under test using the procedure described in paragraphs 4.3.8 through 4.3.10.3. Select the density step nearest the speed point (nearest 1.80 + minimum density) and measure and record this density. Place the test samples in an oven at 150° $\pm$ 2° F for three hours. Remove the samples from the oven. Allow them to cool to room temperature for a few seconds and remeasure the density step measured before the oven treatment. The difference between the density values measured after heating and the density values measured before heating shall be recorded as the loss in density of processed film at the speed point density. Determine the % loss in density.

$$\% \text{ Loss} = \frac{\text{Density Before} - \text{Density After}}{\text{Density Before}} \times 100$$

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. The film shall be packaged in a metal, light-tight water vapor-resistant container of suitable size and design. The metal container shall be adequately sealed against moisture and light with tape conforming to PPP-T-60.

5.1.2 Level C. Films shall be packaged in accordance with the contractor's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Film packaged as specified in 5.1.1 shall be packed in a close fitting box conforming to PPP-B-636, type CF, class weather-resistant, V3c. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required and contain identical quantities. The gross weight of each pack shall be limited to approximately 90 pounds. Container strength and design shall be adequate to insure protection against damage during the shipment, handling, intermediate storage and worldwide redistribution. Containers shall be closed in accordance with the specification appendix. Strapping shall not be required.



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5.2.2 Level B. Film packaged as specified in 5.1.1 shall be packed in a close fitting box conforming to PPP-B-636, type CF, class domestic. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each pack shall be limited to approximately 90 pounds. Container strength and design shall be adequate to insure protection against damage during multiple domestic shipments, handling, and covered storage. Closure of boxes shall be in accordance with method II and the appendix to PPP-B-636.

5.2.3 Level C. Items preserved and packaged as specified in 5.1 shall be packed to comply with the National Motor Freight Classification rules or the Uniform Freight Classification rules.

5.3 Marking. In addition to the marking required by 5.3.3, all marking shall be in accordance with 5.3.1 or 5.3.2 as specified (see 6.2).

5.3.1 Military agencies. All shipping containers shall be marked in accordance with MIL-STD-129, except that the dimensions of the exterior container need not be included. The nomenclature shall be:

FILM PHOTOGRAPHIC, THERMAL DEVELOPING

The type, subtype, size (width and length), expiration date and emulsion or coating numbers shall be clearly marked on the containers.

5.3.2 Civil agencies. The following information shall be included in the labeling provided: Manufacturer's name, trade name of product, width and length, coating or emulsion number, type, subtype and kind, expiration date, exposure index, the name "Film, Photographic, Thermal Developing," perforated or nonperforated, specification number and stock number. Labels shall be attached in such a manner that they will not be damaged when the package is opened. Unless otherwise specified, exterior shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.3 The following precautionary marking shall be permanently and legibly marked in bold contrasting color such as red: "FILM, PHOTOGRAPHIC, THERMAL DEVELOPING", "STORE IN COOL LOCATION".

6. NOTES

6.1 Intended use. The films covered by this specification are intended for use in printing and development equipment of a type recommended by the manufacturer for the reproduction of negative or positive image prints from negative or positive image film masters as shown below.

| From | To |
|---|---|
| a. Negative image | Positive image |
| b. Positive image | Negative image |

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Stock number.
(c) Type, subtype, class and kind.
(d) Width and length.
(e) Selection of samples.
(f) Levels of packing and packaging (see 5.1 and 5.2).
(g) Preproduction samples are or are not required.
(h) Perforated or nonperforated.
(i) Whether civil or military marking is required (see 5.3).



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6.3 Guarantee. Upon request, the manufacturer shall replace without cost to the Government, any material which exhibits latent or other physical defects prior to the expiration date.

6.4 Cross reference of classification. Paragraph 1.2.1 of the superseded basic specification has been completely revised in this specification. The items covered by this specification may be binned with prior items. The difference in classification between the basic superseded specification and this specification are shown in the cross reference below:

| Classification | Basic specification | This specification |
|---|---|---|
| Type | Heat developing | Type I - Duplicating |
| Subtypes | None | A - Nonreversible |
| Classes | A - 2 mil base<br>B - 3 mil base<br>C - 5 mil base | 1 - High speed |
| Kinds | None | 1A - 2 mil<br>1B - 3 mil<br>1C - 4 mil<br>1D - 5 mil<br>1E - 7 mil |

6.5 Definitions.

6.5.1 Kalvar Standard Sensitometer, Model B. A doped lamp sensitometer designed and developed for precise and reproducible measurements of the ultraviolet-light sensitivity of vesicular film materials (see 4.3.7, items # 1 and # 11).

6.5.2 Light scattering. This term is used to describe the method of image formation in films in which opacity is obtained by the creation of light scattering elements which reflect and diffract light incident upon the film as opposed to those films in which the incident light is attenuated by absorption. Such light scattering films are characterized by a very marked difference between the visual diffuse transmission density, the printing diffuse transmission density and the projection density. The printing diffuse transmission density range and the projection density range are much greater than the visual diffuse transmission density range.

6.5.3 Projection density f/4.5. Is defined to be the common logarithm of the ratio of radiant flux incident normal (or deviating from normal no more than 6.3°) to the radiant flux collected by a f/4.5 aperature located in the forward scattering direction. Projection density f/4.5 values are obtained from an absolute projection densitometer f/4.5, such as Kalvar's - Baus Projection Densitometer or its equivalent.

6.5.4 Maximum density. Dmax is the projection density obtained when a light scattering film is exposed sufficiently to decompose all of the sensitizer and when this piece of film is processed according to the manufacturer's instructions.

6.5.5 Vesicle. A nearly spherical body usually less than one micron in size constituting a thin-walled cavity in a film which acts as a scattering site for the light transmitted through the film.

6.5.6 Vesicular. Of relating to vesicles; or having from the structure of vesicle; containing vesicles; composed of vesicles; characterized by vesicles.



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6.5.7 Clearing. Is defined as the act by which the suitable application of ultraviolet light desensitizes the residual sensitizer present in the film after exposure and development.

Military Custodians:

Army - EL
Navy - AS
Air Force - 70

Review Activities:

Army - EL
Navy - AS, OS, MC
Air Force - 70

User Activities:

Army - EL
Navy - AS, OS, MC

Preparing Activity:

GSA - FSS

Civil Agency Coordinating Activities:

HEW - SSA
Treasury - IRS





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